



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: / /

Laboratory Manual

Academic Year	:	2024-25	Semester	:	4
Course code	:	CSE206	Course name	:	DATABASE MANAGEMENT SYSTEM

Practical - 10

Aim: - You are a database developer for a financial application that requires frequent mathematical calculations. One critical operation is multiplying two numbers provided by users and returning the result. To achieve this, you must create a robust PL/SQL procedure that handles various scenarios, including positive and negative numbers, zero values, and floating-point numbers. Create a PL/SQL procedure that calculates the product of two numbers, ensuring error handling for invalid inputs and consistent results.

Tasks:-

1. A user wants to calculate the product of two positive numbers, e.g., 10 and 5, to determine total costs.

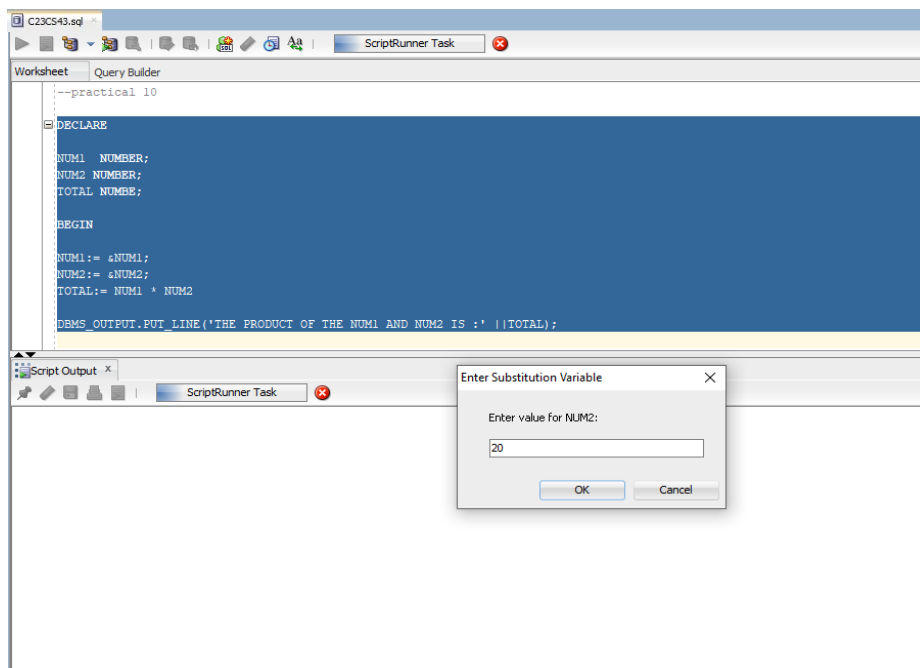
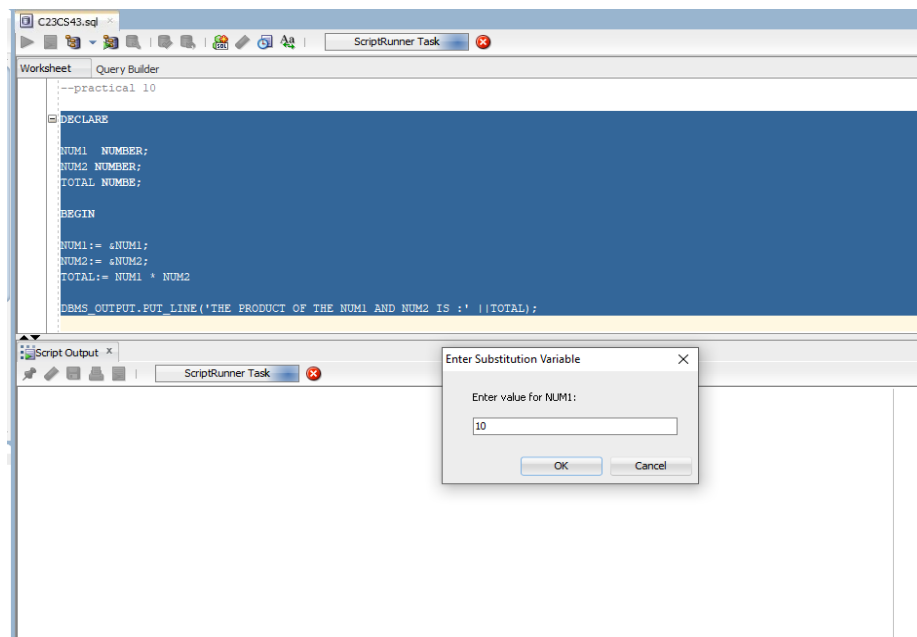
```

C23CS043.sql
Worksheet  Query Builder
FROM CUSTOMER2 c;

--practical 10

DECLARE
    NUM1 NUMBER;
    NUM2 NUMBER;
    TOTAL NUMBER;
BEGIN
    NUM1:= 60;
    NUM2:= 10;
    TOTAL:= NUM1 * NUM2;
    DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

Script Output x
Task completed in 0.06 seconds
NUM2:= 50;
TOTAL:= NUM1 * NUM2;
DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;
PL/SQL procedure successfully completed.
PL/SQL procedure successfully completed.
  
```



```

C23CS43.sql
Worksheet Query Builder
FROM CUSTOMER2 c;

--practical 10

DECLARE
NUM1 NUMBER;
NUM2 NUMBER;
TOTAL NUMBER;

BEGIN
NUM1:= &NUM1;
NUM2:= &NUM2;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

Script Output x
Task completed in 8.301 seconds

BEGIN

NUM1:= 20;
NUM2:= 10;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

PL/SQL procedure successfully completed.

```

```

C23CS23
Worksheet Query Builder
0.29899999 seconds

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

CREATE OR REPLACE PROCEDURE multiptiption(num1 IN NUMBER , num2 IN NUMBER)
AS
BEGIN
dbms_output.put_line('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' || NUM1 * NUM2);
END;

DECLARE
INP_1 NUMBER := 23;
INP_2 NUMBER :=23;
BEGIN
MULTIPLICATION(INP_1,INP_2);
END;

Script Output x
Task completed in 0.299 seconds

BEGIN

NUM1:= 20;
NUM2:= 10;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

PL/SQL procedure successfully completed.

Procedure MUPLTIPICTION compiled

```

```

C23CS43.sql
Worksheet Query Builder
0.041 seconds

NUM2:= &NUM2;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

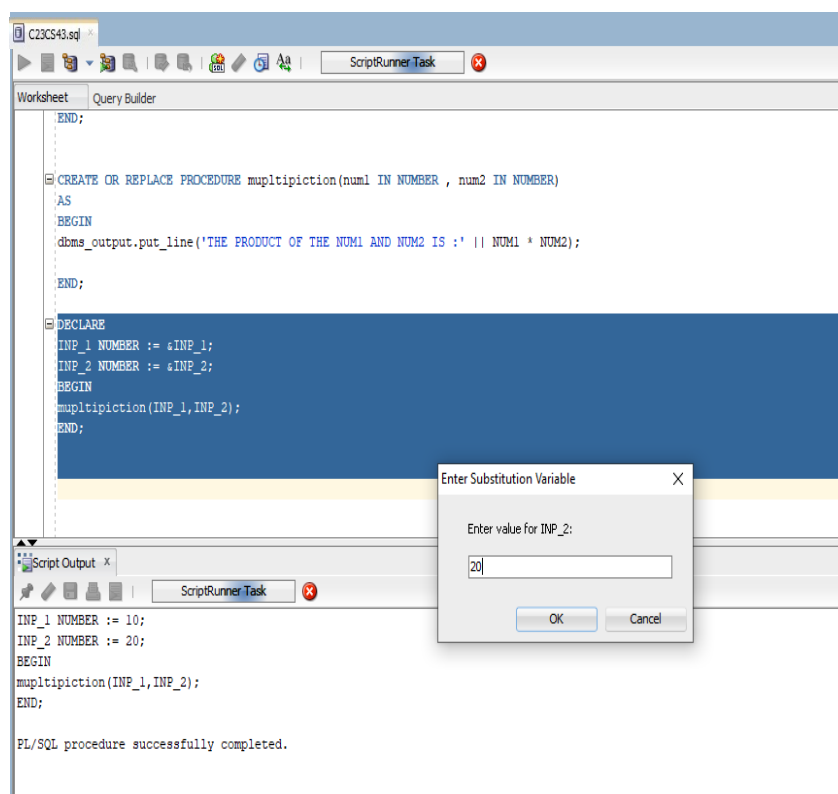
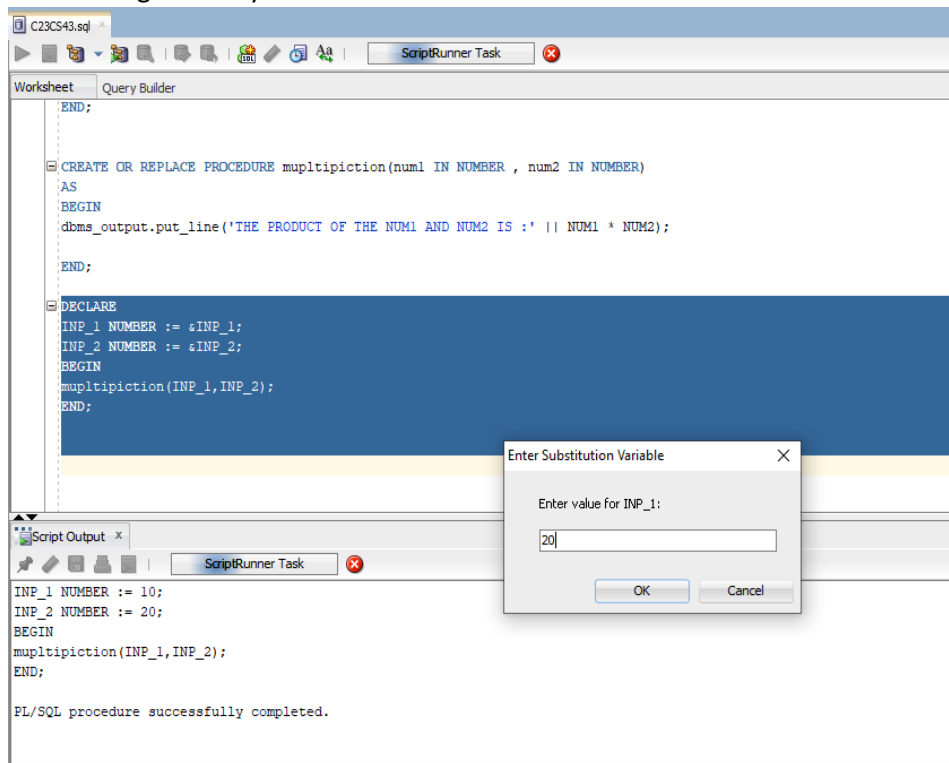
CREATE OR REPLACE PROCEDURE multiptiption(num1 IN NUMBER , num2 IN NUMBER)
AS
BEGIN
dbms_output.put_line('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' || NUM1 * NUM2);
END;

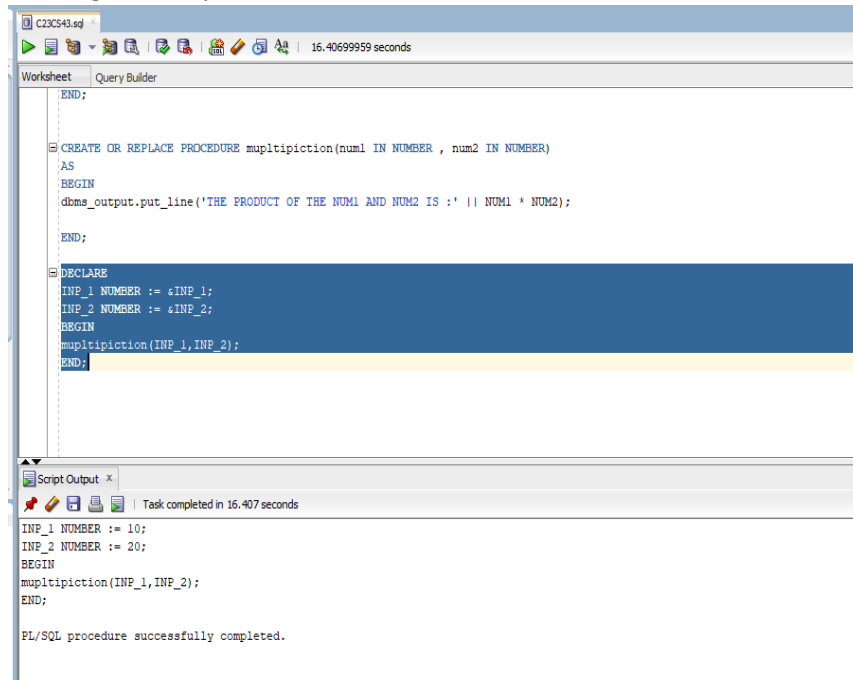
DECLARE
INP_1 NUMBER := 23;
INP_2 NUMBER :=23;
BEGIN
multiptiption(INP_1,INP_2);
END;

Script Output x
Task completed in 0.041 seconds

BEGIN
MULTIPLICATION(INP_1,INP_2);
END;
Error report -
ORA-06550: line 5, column 1:
PLS-00201: identifier 'MULTIPLICATION' must be declared
ORA-06550: line 5, column 1:
PL/SQL: Statement ignored
06550. 00000 - "line %s, column %s:\n%s"

```





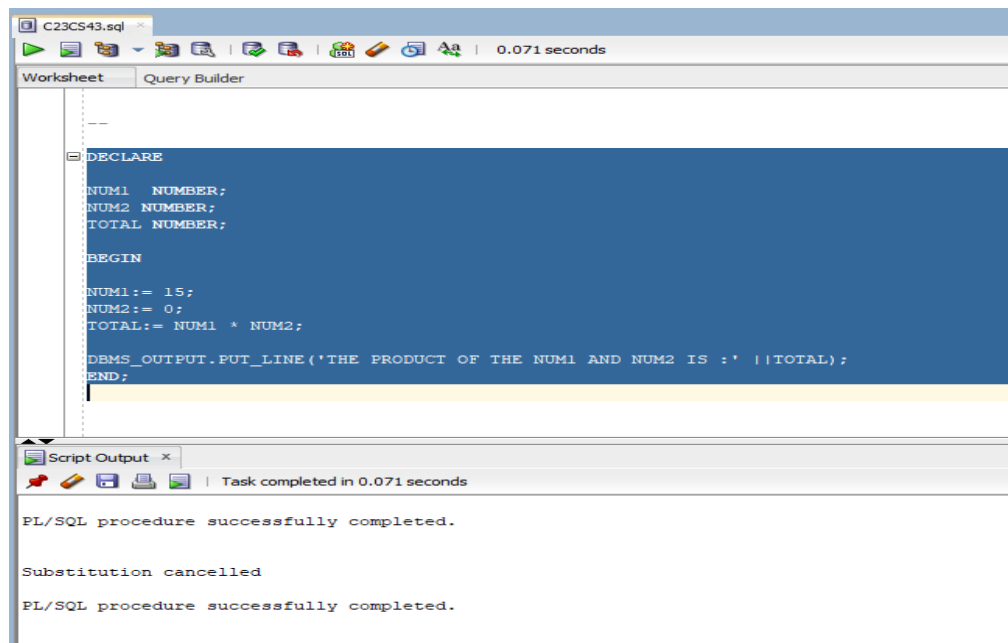
The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS43.sql'. The main editor contains a PL/SQL procedure named 'mulptipiction' (note the typo) that takes two numbers as input and outputs their product. The procedure is defined as follows:

```
END;  
  
CREATE OR REPLACE PROCEDURE mulptipiction(num1 IN NUMBER , num2 IN NUMBER)  
AS  
BEGIN  
  dbms_output.put_line('THE PRODUCT OF THE NUM1 AND NUM2 IS :' || NUM1 * NUM2);  
  
END;  
  
DECLARE  
  INP_1 NUMBER := 10;  
  INP_2 NUMBER := 20;  
BEGIN  
  mulptipiction(INP_1,INP_2);  
END;
```

The 'Script Output' window at the bottom shows the execution results:

```
INP_1 NUMBER := 10;  
INP_2 NUMBER := 20;  
BEGIN  
mulptipiction(INP_1,INP_2);  
END;  
  
PL/SQL procedure successfully completed.
```

2. A user wants to verify that multiplying any number with zero results in zero, e.g., 15 and 0.

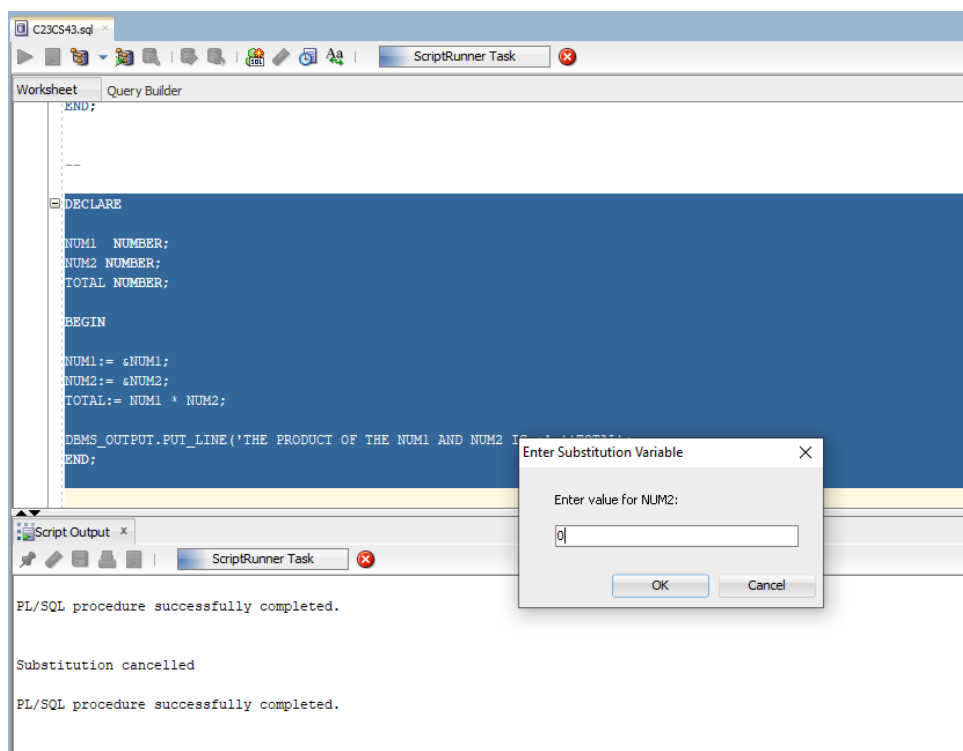
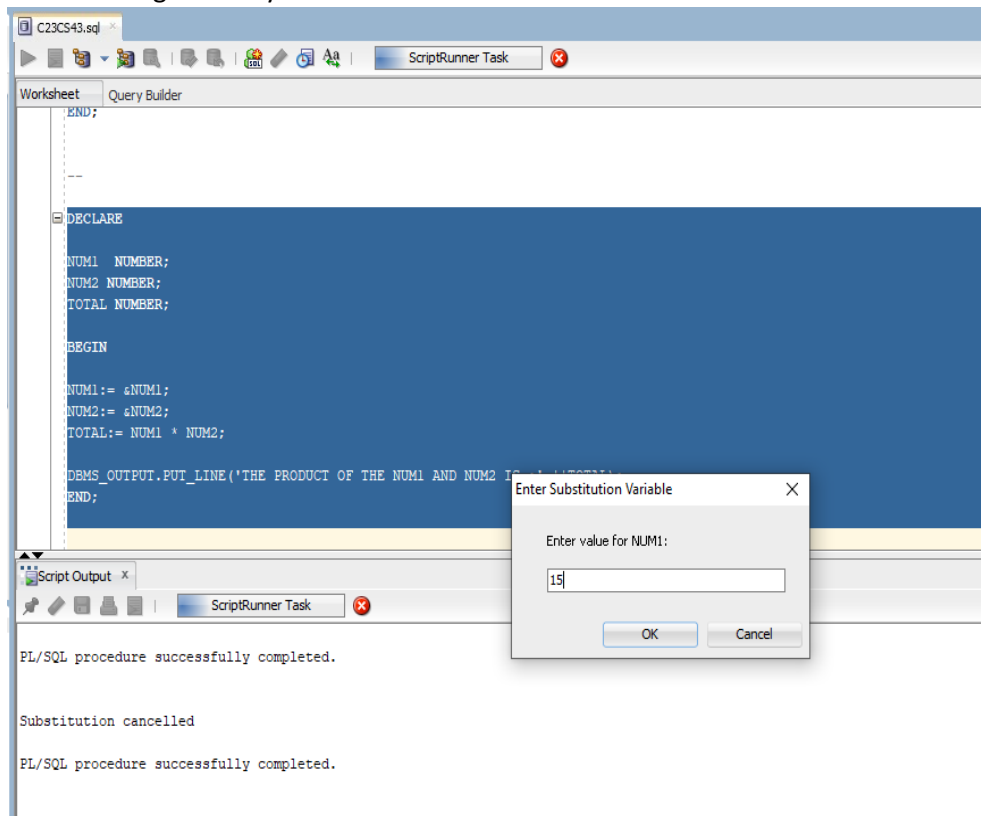


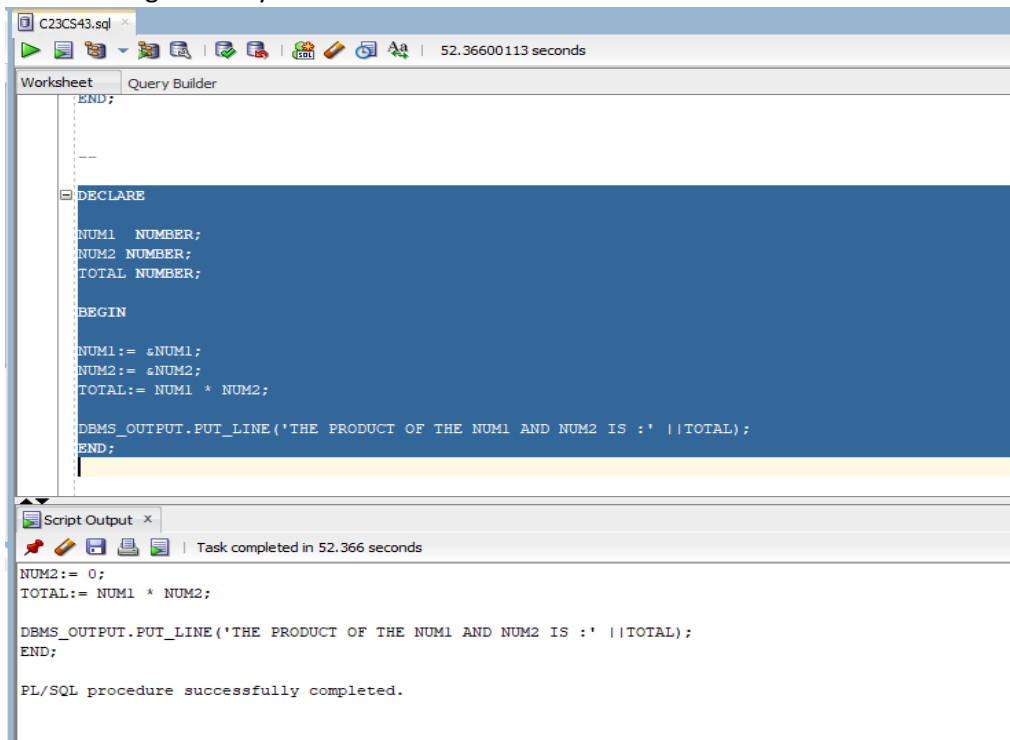
The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS43.sql'. The main editor contains a PL/SQL procedure that declares three variables: NUM1, NUM2, and TOTAL. It sets NUM1 to 15 and NUM2 to 0, then calculates the product and outputs the result.

```
--  
  
DECLARE  
  NUM1 NUMBER;  
  NUM2 NUMBER;  
  TOTAL NUMBER;  
  
BEGIN  
  NUM1:= 15;  
  NUM2:= 0;  
  TOTAL:= NUM1 * NUM2;  
  
  DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS :' ||TOTAL);  
END;
```

The 'Script Output' window at the bottom shows the execution results:

```
PL/SQL procedure successfully completed.  
  
Substitution cancelled  
PL/SQL procedure successfully completed.
```





The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS43.sql'. The 'Query Builder' tab is active. The SQL code in the worksheet is as follows:

```
END;

--

DECLARE

NUM1 NUMBER;
NUM2 NUMBER;
TOTAL NUMBER;

BEGIN

NUM1:= &NUM1;
NUM2:= &NUM2;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);

END;
```

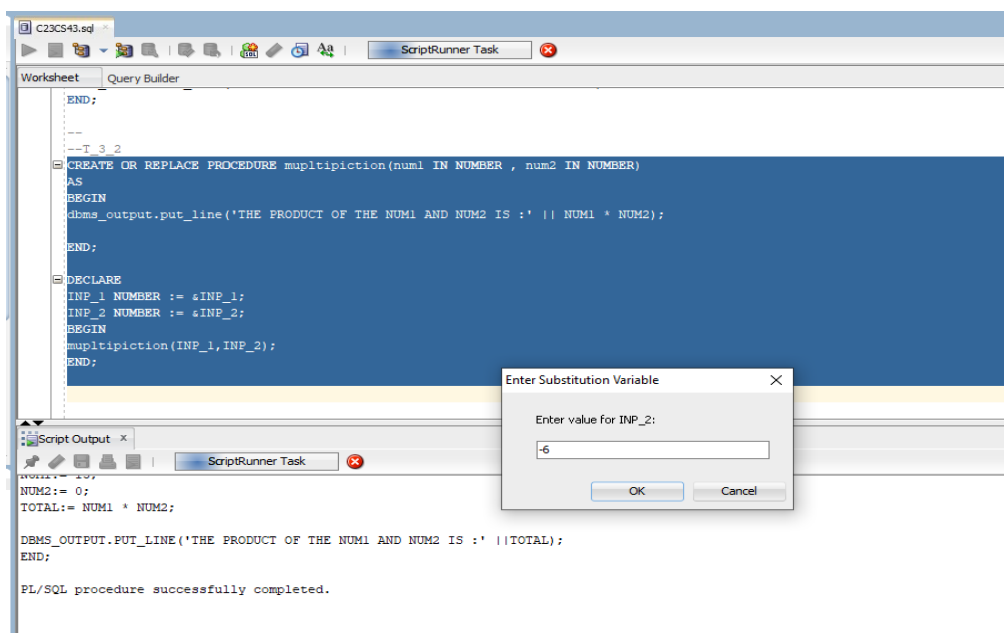
The 'Script Output' window at the bottom shows the execution results:

```
NUM2:= 0;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

PL/SQL procedure successfully completed.
```

3. A user is calculating the product of two negative numbers, e.g., -4 and -6, which should result in a positive product.



The screenshot shows the SQL Developer interface with a worksheet titled 'C23CS43.sql'. The 'Query Builder' tab is active. The SQL code in the worksheet is as follows:

```
END;

--T 3 2

CREATE OR REPLACE PROCEDURE mupltipiction(num1 IN NUMBER , num2 IN NUMBER)
AS
BEGIN
dbms_output.put_line('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' || NUM1 * NUM2);
END;

DECLARE
INP_1 NUMBER := &INP_1;
INP_2 NUMBER := &INP_2;
BEGIN
mupltipiction(INP_1, INP_2);
END;
```

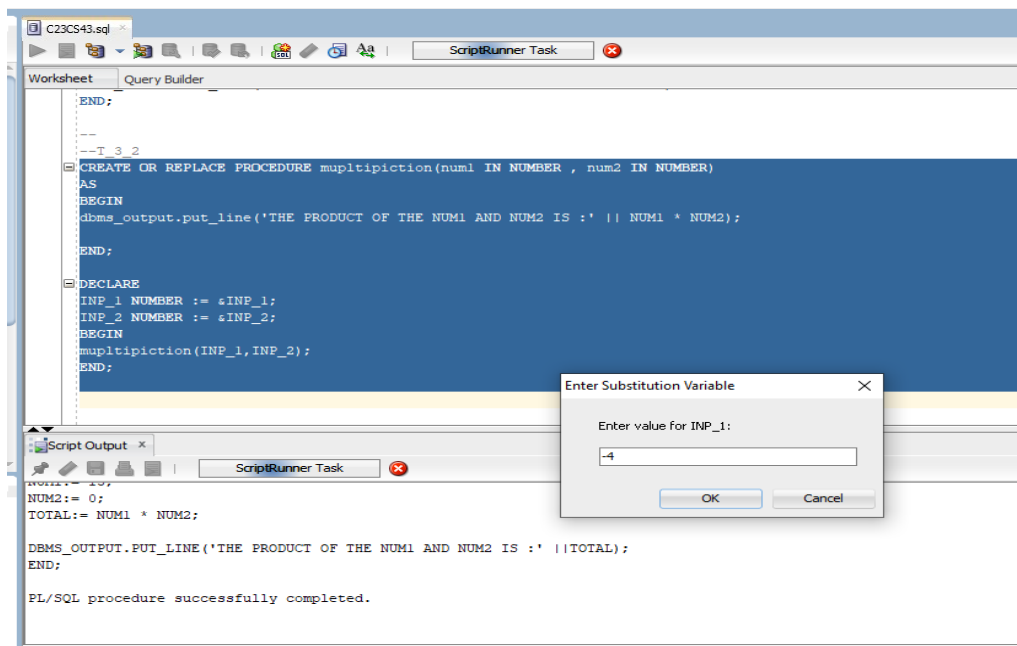
The 'Script Output' window at the bottom shows the execution results:

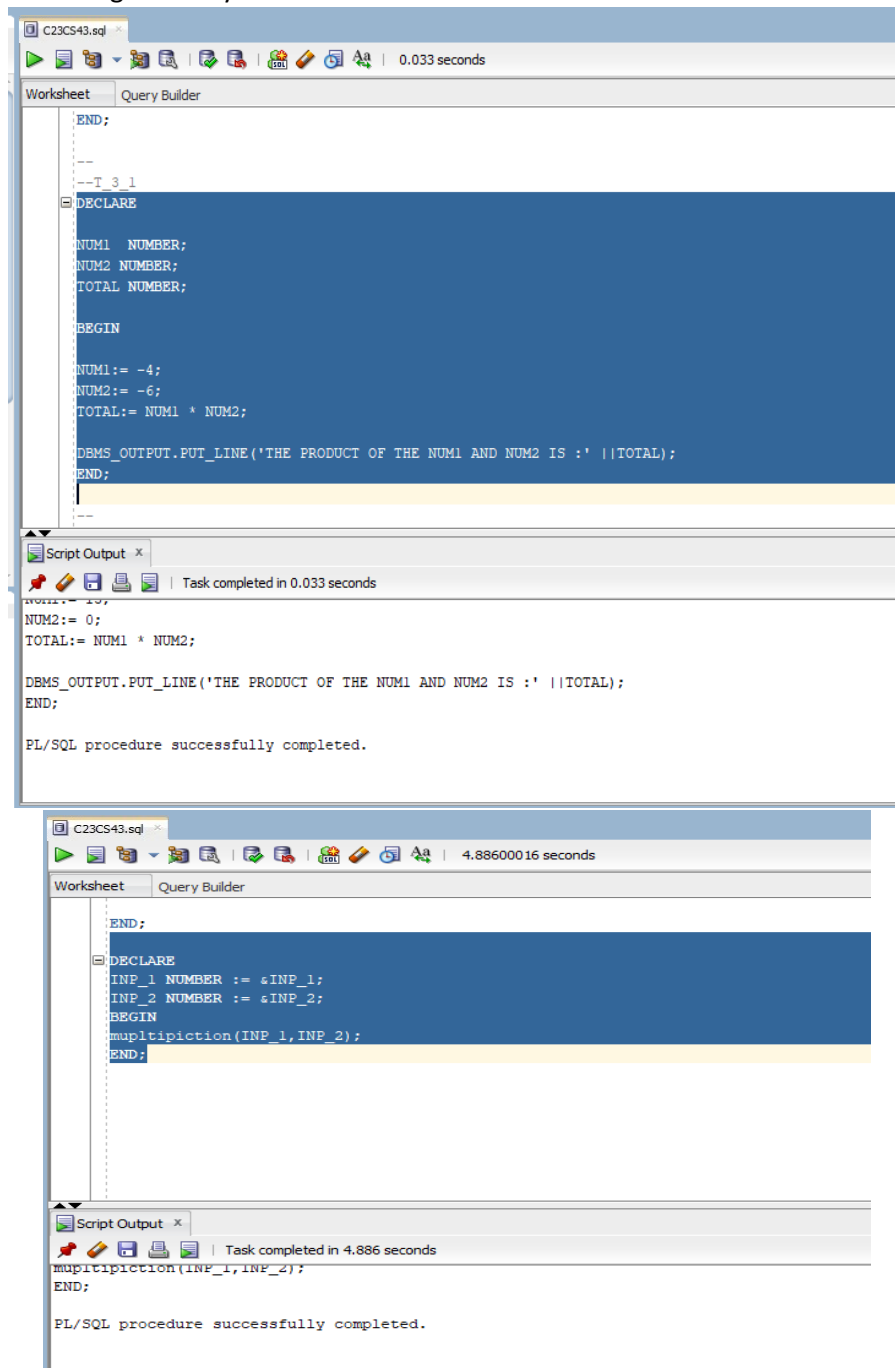
```
NUM2:= 0;
TOTAL:= NUM1 * NUM2;

DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

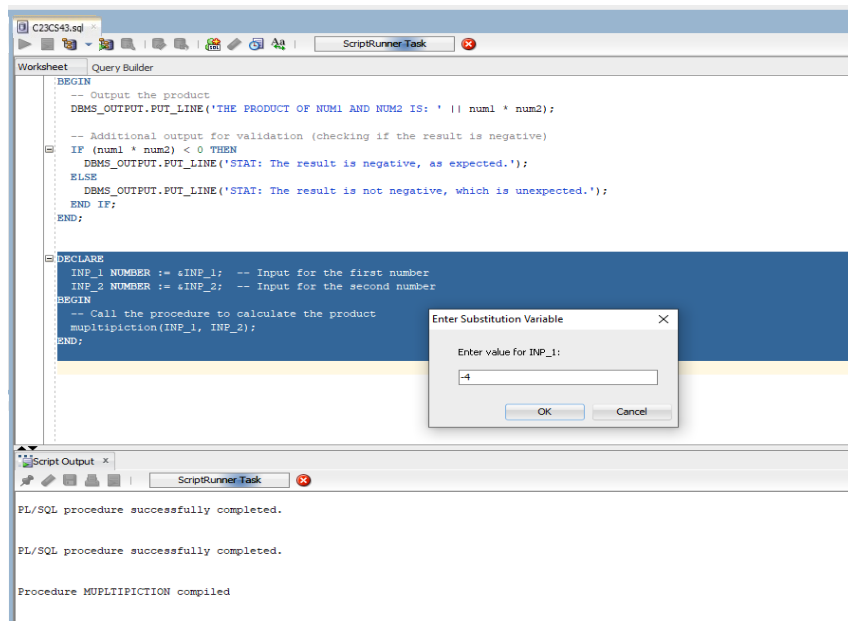
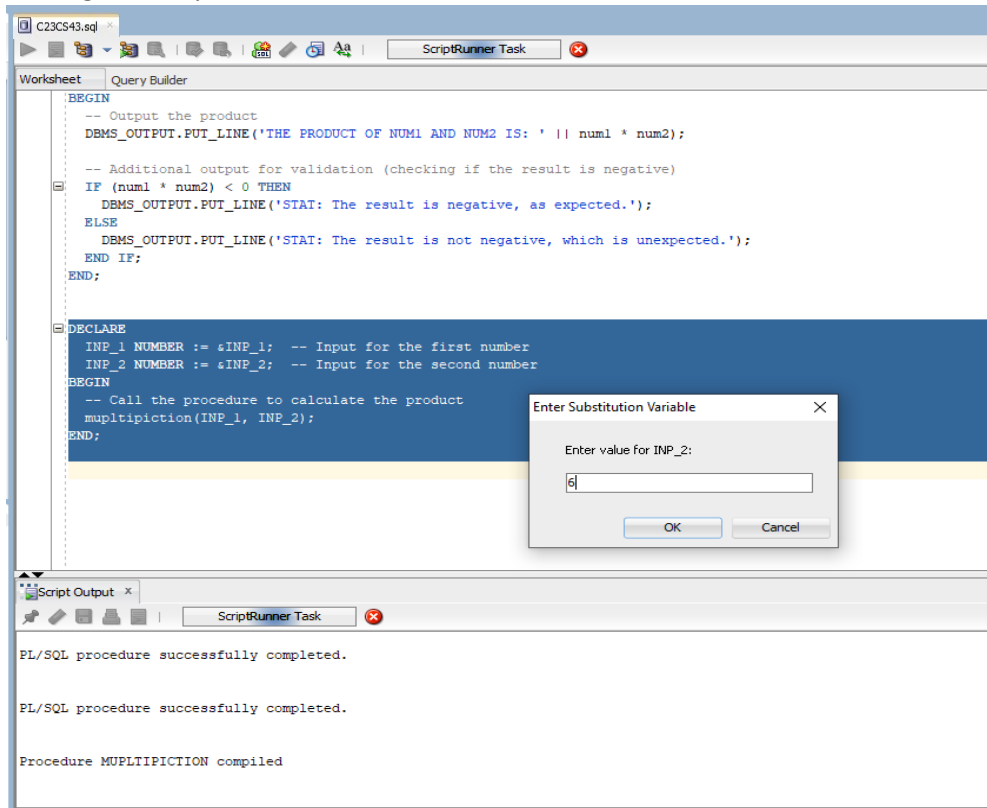
PL/SQL procedure successfully completed.
```

An 'Enter Substitution Variable' dialog box is open, prompting for the value of 'INP_2'. The input field contains '-6'.





4. A user is calculating the product of a positive and a negative number, e.g., 7 and -3, to validate the outcome as negative.



The image shows two screenshots of the SQL Developer interface. The top screenshot displays a PL/SQL procedure named `multipiction` (note the typo) that takes two numbers as input and returns their product. It includes a validation step to check if the result is negative. The bottom screenshot shows a PL/SQL block that declares variables `NUM1`, `NUM2`, and `TOTAL`, assigns values to `NUM1` and `NUM2`, calculates the product, and outputs the result with a validation message.

Top Screenshot:

```
END;

--

CREATE OR REPLACE PROCEDURE multipiction(num1 IN NUMBER, num2 IN NUMBER)
AS
BEGIN
    -- Output the product
    DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF NUM1 AND NUM2 IS: ' || num1 * num2);

    -- Additional output for validation (checking if the result is negative)
    IF (num1 * num2) < 0 THEN
        DBMS_OUTPUT.PUT_LINE('STAT: The result is negative, as expected.');

```
END IF;
END;

DECLARE
 INP_1 NUMBER := &INP_1; -- Input for the first number
 INP_2 NUMBER := &INP_2; -- Input for the second number
BEGIN
 -- Call the procedure to calculate the product
 multipiction(INP_1, INP_2);
END;
```



Script Output:



```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

Procedure MUPLTIPICTION compiled
```



Bottom Screenshot:



```
DECLARE
 NUM1 NUMBER;
 NUM2 NUMBER;
 TOTAL NUMBER;
BEGIN
 -- Example: Multiplying a positive and negative number
 NUM1 := 7;
 NUM2 := -3;
 TOTAL := NUM1 * NUM2;

 -- Output the product
 DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF NUM1 AND NUM2 IS: ' || TOTAL);

 -- Additional output for validation
 IF TOTAL < 0 THEN
 DBMS_OUTPUT.PUT_LINE('STAT: The result is negative, as expected.');

```
END IF;
END;
```

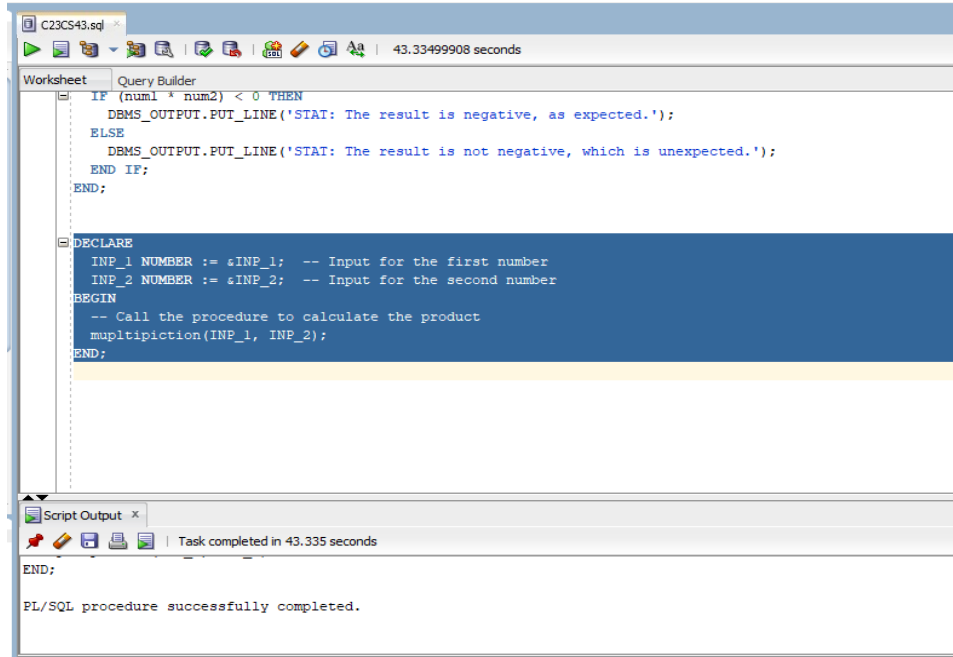

Script Output:


```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.
```


```


```



The screenshot displays the Oracle SQL Developer environment. The top window, titled 'C23CS43.sql', is the 'Query Builder' tab, showing a PL/SQL procedure. The procedure includes an IF statement to check if the product of two numbers is negative, followed by a DECLARE section for input variables, and a BEGIN section that calls a procedure named 'multiplication'. The bottom window, titled 'Script Output', shows the execution results, indicating that the task was completed successfully in 43.335 seconds.

```
IF (num1 * num2) < 0 THEN
  DBMS_OUTPUT.PUT_LINE('STAT: The result is negative, as expected.');
```

```
ELSE
  DBMS_OUTPUT.PUT_LINE('STAT: The result is not negative, which is unexpected.');
```

```
END IF;
```

```
END;
```

```
DECLARE
  INP_1 NUMBER := &INP_1; -- Input for the first number
  INP_2 NUMBER := &INP_2; -- Input for the second number
BEGIN
  -- Call the procedure to calculate the product
  multiplication(INP_1, INP_2);
END;
```

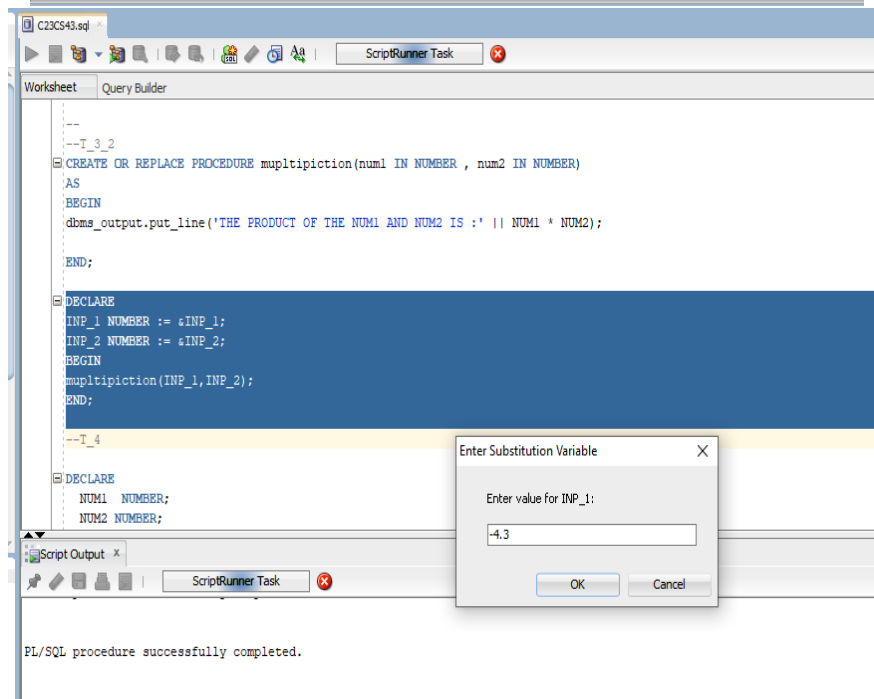
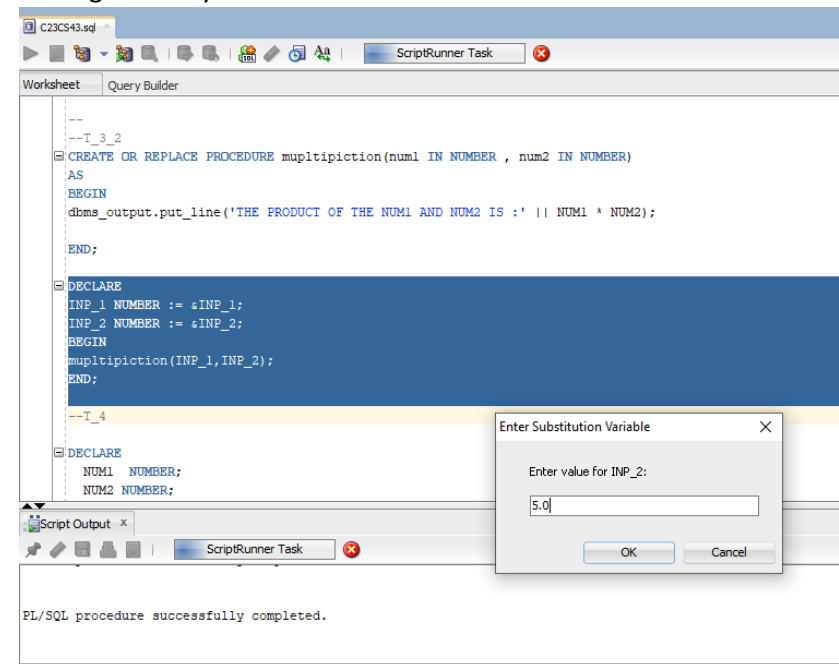
Script Output x

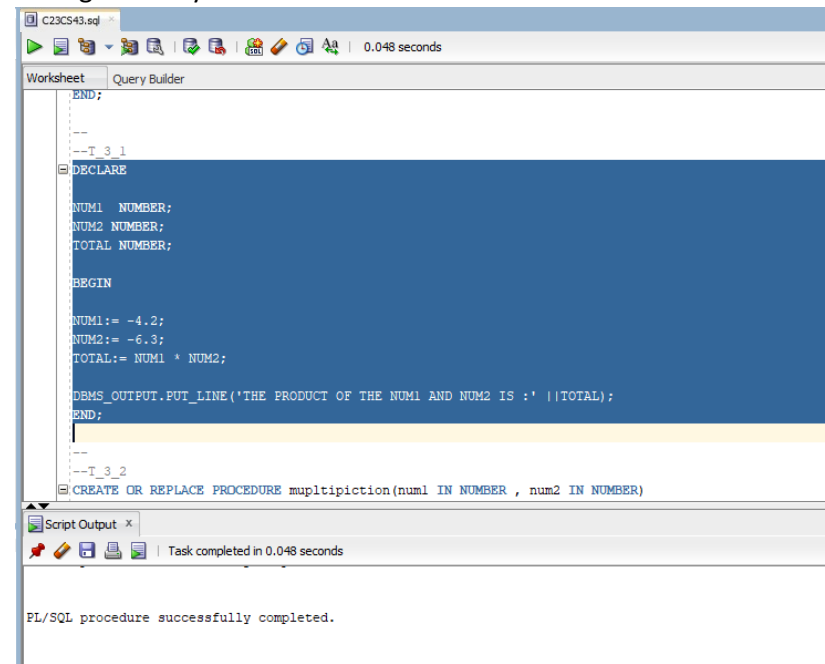
Task completed in 43.335 seconds

END;

PL/SQL procedure successfully completed.

5. A user wants to calculate the product of two decimal numbers, e.g., 2.5 and 4.2, for precision-sensitive operations.





```
END;

--T_3_1
DECLARE
    NUM1 NUMBER;
    NUM2 NUMBER;
    TOTAL NUMBER;

BEGIN
    NUM1:= -4.2;
    NUM2:= -6.3;
    TOTAL:= NUM1 * NUM2;

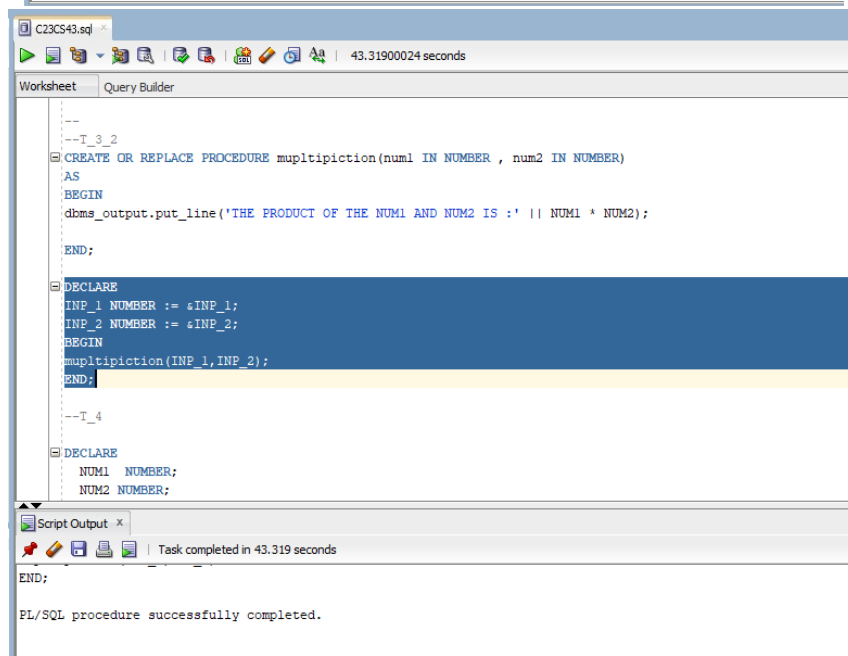
    DBMS_OUTPUT.PUT_LINE('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' ||TOTAL);
END;

--T_3_2
CREATE OR REPLACE PROCEDURE mupltipiction(num1 IN NUMBER , num2 IN NUMBER)
```

Script Output x

Task completed in 0.048 seconds

PL/SQL procedure successfully completed.



```
--T_3_2
CREATE OR REPLACE PROCEDURE mupltipiction(num1 IN NUMBER , num2 IN NUMBER)
AS
BEGIN
    dbms_output.put_line('THE PRODUCT OF THE NUM1 AND NUM2 IS : ' || NUM1 * NUM2);

END;

DECLARE
    INP_1 NUMBER := &INP_1;
    INP_2 NUMBER := &INP_2;
BEGIN
    mupltipiction(INP_1,INP_2);
END;

--T_4
DECLARE
    NUM1 NUMBER;
    NUM2 NUMBER;
```

Script Output x

Task completed in 43.319 seconds

END;

PL/SQL procedure successfully completed.

EXTRA QUERY:-

The screenshot shows a SQL Developer window with a PL/SQL procedure named 'C23CS43.sql'. The procedure is designed to calculate the product of two input numbers and then select data from the 'EMPLOYEE' table. However, it has encountered a syntax error at line 18, column 3, where a semicolon was used instead of an 'end-of-file' symbol to continue the statement.

```

BEGIN
    -- Call the procedure to calculate the product
    mupltipiction(INP_1, INP_2);
END;

--EXTRA

SELECT * FROM EMPLOYEE

DECLARE
    EMP_NAME VARCHAR (50);
    EMP_SAL NUMBER;
BEGIN
    SELECT EMP_NAME,EMP_SAL INTO EMP_NAME,EMP_SAL
    FROM EMPLOYEE
    WHERE EMP_ID = 104;

    DBMS_OUTPUT.PUT_LINE('EMPLOYEE NAME :'||EMP_NAME);
    DBMS_OUTPUT.PUT_LINE('EMPLOYEE SALARY: '||EMP_SAL);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('NO DATA FOUND');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('ERROR: ' || SQLERRM);
END;

```

Error Report:
 Error report -
 ORA-06550: line 18, column 3:
 PL/SQL: Encountered the symbol "end-of-file" when expecting one of the following:

 ; <an identifier> <a double-quoted delimited-identifier>
 The symbol ":" was substituted for "end-of-file" to continue.
 06550. 00000 - "line %s, column %s:\n%s"
 *Cause: Usually a PL/SQL compilation error.
 *Action:

 PL/SQL procedure successfully completed.

The screenshot shows the same SQL Developer window after a successful query execution. The 'Query Result' tab is active, displaying a table with 6 rows of employee data. The 'Script Output' tab is also visible, showing the execution of the PL/SQL procedure.

Query Result:

EMP_ID	EMP_NAME	EMP_SAL	JOB_ID	DEPT_NO	PHONE
102	Adama	2500	MGR_MGR	10	2345678901
103	Anita	5000	IT_PROG	30	3456789012
104	Anamika	2975	LEC	15	4567890123
105	Snehal	1600	COMP_OP	25	(null)
12	BANTI	43222.58		10	1234567890

Script Output:

```

INP_1 NUMBER := &INP_1; -- Input for the first number
INP_2 NUMBER := &INP_2; -- Input for the second number
BEGIN
    -- Call the procedure to calculate the product
    mupltipiction(INP_1, INP_2);
END;

--EXTRA

SELECT * FROM EMPLOYEE

DECLARE
    EMP_ID NUMBER;
    EMP_NAME VARCHAR(50);
    EMP_SAL NUMBER;
    JOB_ID VARCHAR(30);
    PHONE VARCHAR(30);

```